



Experiment 1

Private Cloud using OpenStack

AIM:

To set up a local private cloud environment on a single Ubuntu machine using OpenStack (MicroStack). This lab gives hands-on experience in managing cloud resources like compute instances, virtual networks, and storage space.

What is Openstack?

OpenStack is a popular open-source software platform designed to set up and manage cloud infrastructure. It operates as Infrastructure as a Service (IaaS), allowing users to create, run, and scale virtual machines on demand while handling processing, network connections, and data storage.

What is Microstack?

MicroStack is an easy-to-use, single-node distribution of OpenStack delivered via a Snap package. It simplifies setup by handling complex dependencies automatically, making it ideal for testing, learning, and small local developments without needing a multi-server setup.

What is Private Cloud?

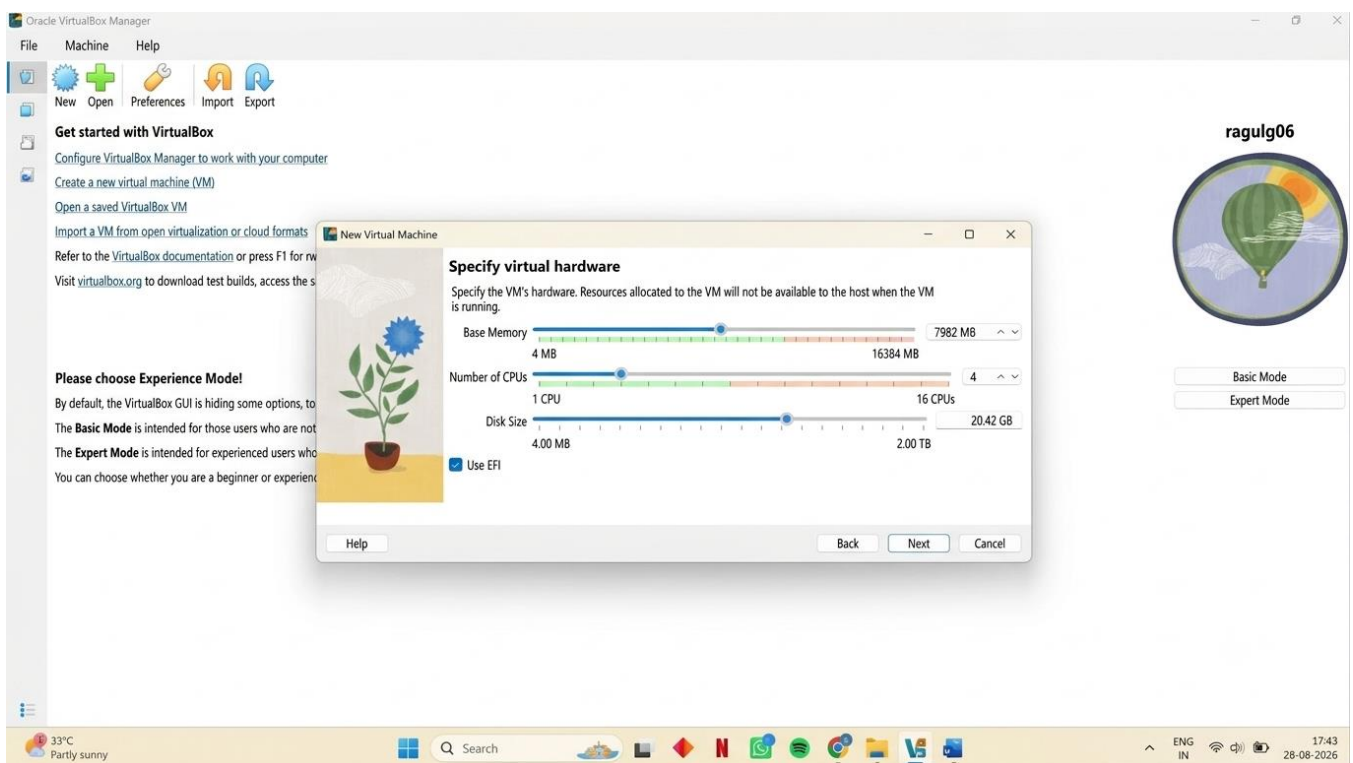
A private cloud is a dedicated cloud computing setup reserved exclusively for a single organization or user. Unlike public clouds, it offers dedicated resources, enhanced control over security, and customizable configurations within an isolated environment.

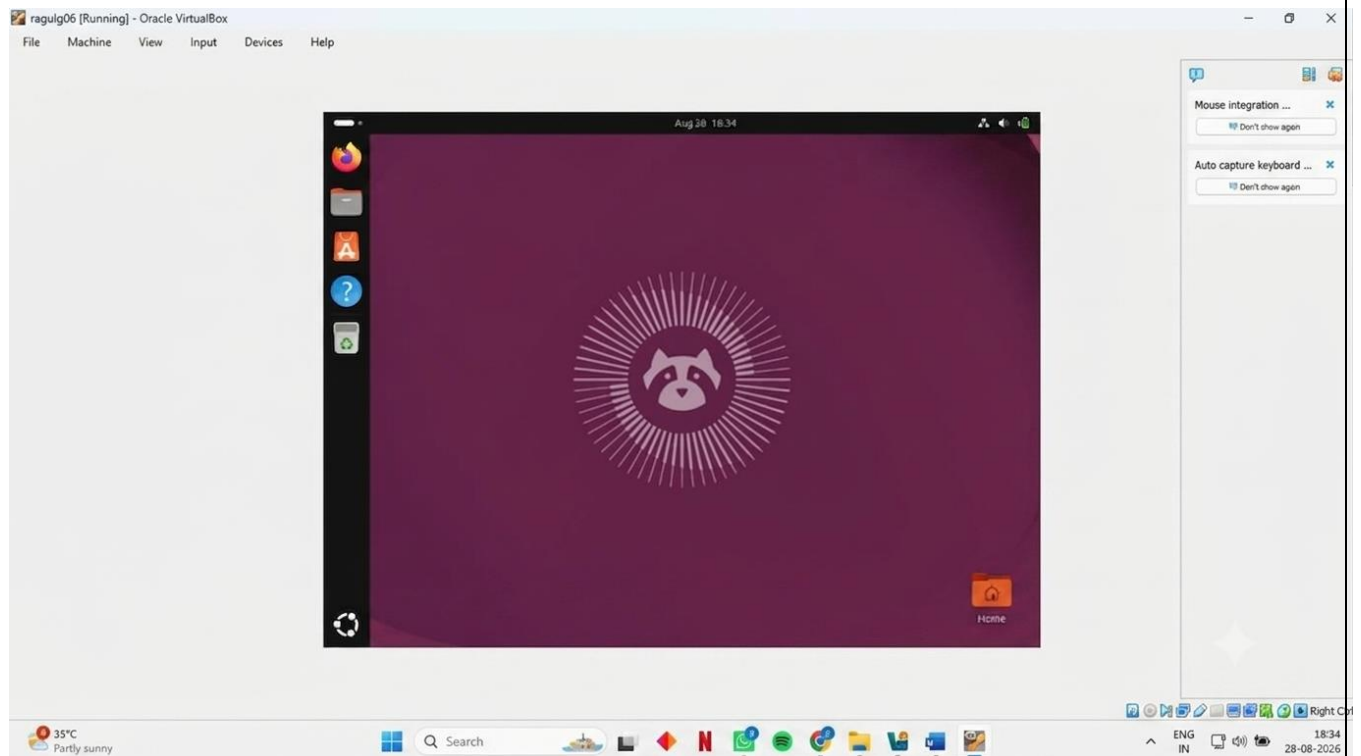
PROCEDURE:

Step 1

Create a new Ubuntu virtual machine using Oracle VirtualBox with the following configuration:

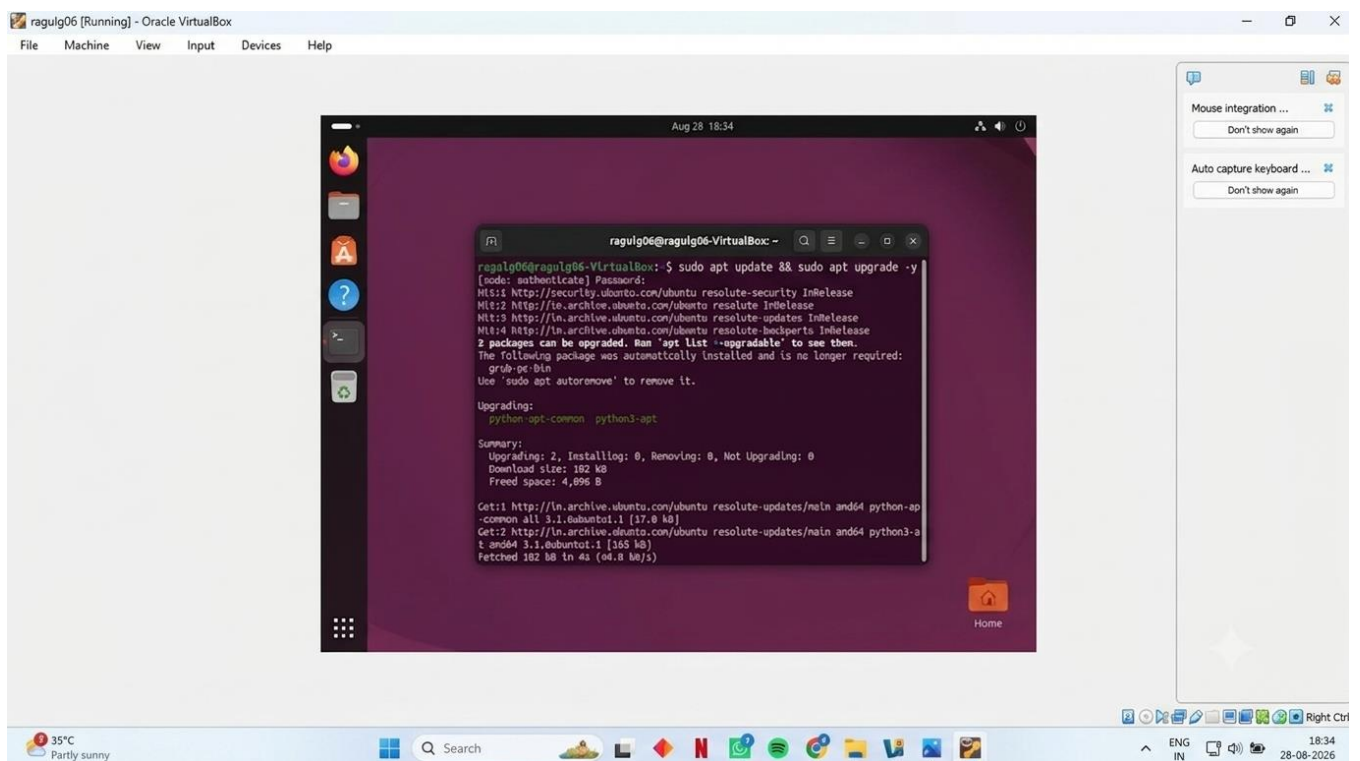
- **Memory:** 8 GB RAM
- **Processor:** 4 CPU Cores
- **Storage:** 20 GB Virtual Disk





Step 2

Install the Ubuntu operating system and log in using the username and password created during installation.



Step 3

Update the Ubuntu packages: `sudo apt update`
`sudo apt upgrade`

Step 4

Install MicroStack (Beta channel): `sudo snap install microstack --beta`

Step 5

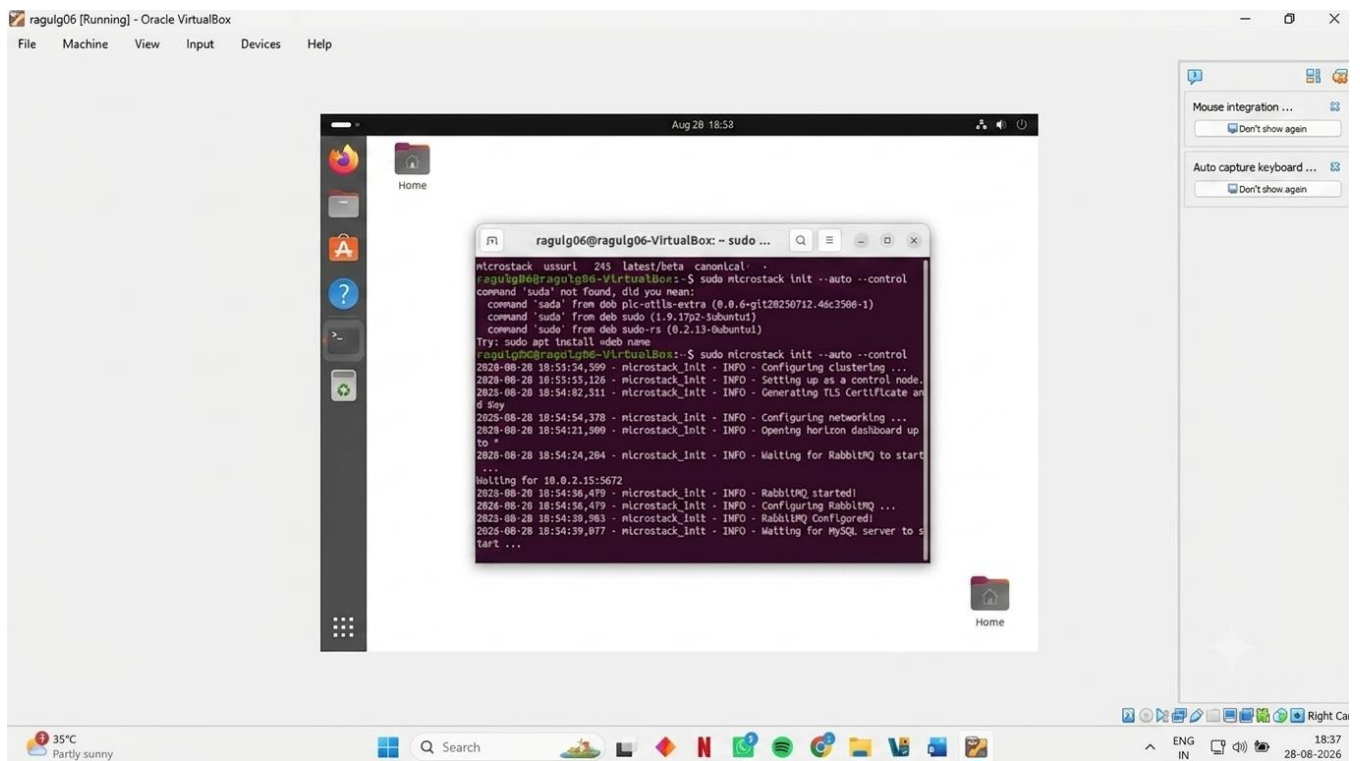
Check whether MicroStack has been installed successfully.

`snap list microstack`

Step 6

Initialize the OpenStack services automatically by executing the following command.

`sudo microstack init --auto --control`



The screenshot shows a VirtualBox window titled 'ragulg06 [Running] - Oracle VirtualBox'. Inside the VM, a terminal window is open with the following output:

```
ragulg06@ragulg06-VirtualBox: ~ sudo ...
microstack ussurl 245 latest/beta canonical' -
ragulg06@ragulg06-VirtualBox:~$ sudo microstack init --auto --control
command 'sudo' not found, did you mean:
  command 'sada' from deb plc-utils-extra (0.0.6-git20250712.44c3506-1)
  command 'sude' from deb sudo (1.9.1702-subuntui)
  command 'sude' from deb sudo-rs (0.2.13-0ubuntu1)
Try: sudo apt install sdeb name
ragulg06@ragulg06-VirtualBox:~$ sudo microstack init --auto --control
2025-08-20 10:53:54,599 - microstack_init - INFO - Configuring clustering ...
2025-08-20 10:53:55,126 - microstack_init - INFO - Setting up as a control node.
2025-08-20 10:54:02,511 - microstack_init - INFO - Generating TLS Certificate an
d key
2025-08-20 10:54:54,370 - microstack_init - INFO - Configuring networking ...
2025-08-20 10:54:21,999 - microstack_init - INFO - Opening horizon dashboard up
to *
2025-08-20 10:54:24,284 - microstack_init - INFO - Waiting for RabbitMQ to start
...
Waiting for 10.0.2.15:5672
2025-08-20 10:54:36,479 - microstack_init - INFO - RabbitMQ started!
2025-08-20 10:54:56,479 - microstack_init - INFO - Configuring RabbitMQ ...
2025-08-20 10:54:39,963 - microstack_init - INFO - RabbitMQ Configured!
2025-08-20 10:54:39,077 - microstack_init - INFO - Waiting for MySQL server to s
tart ...
```

Step 7

Verify the installed OpenStack version:

```
microstack.openstack --version
```

Step 8

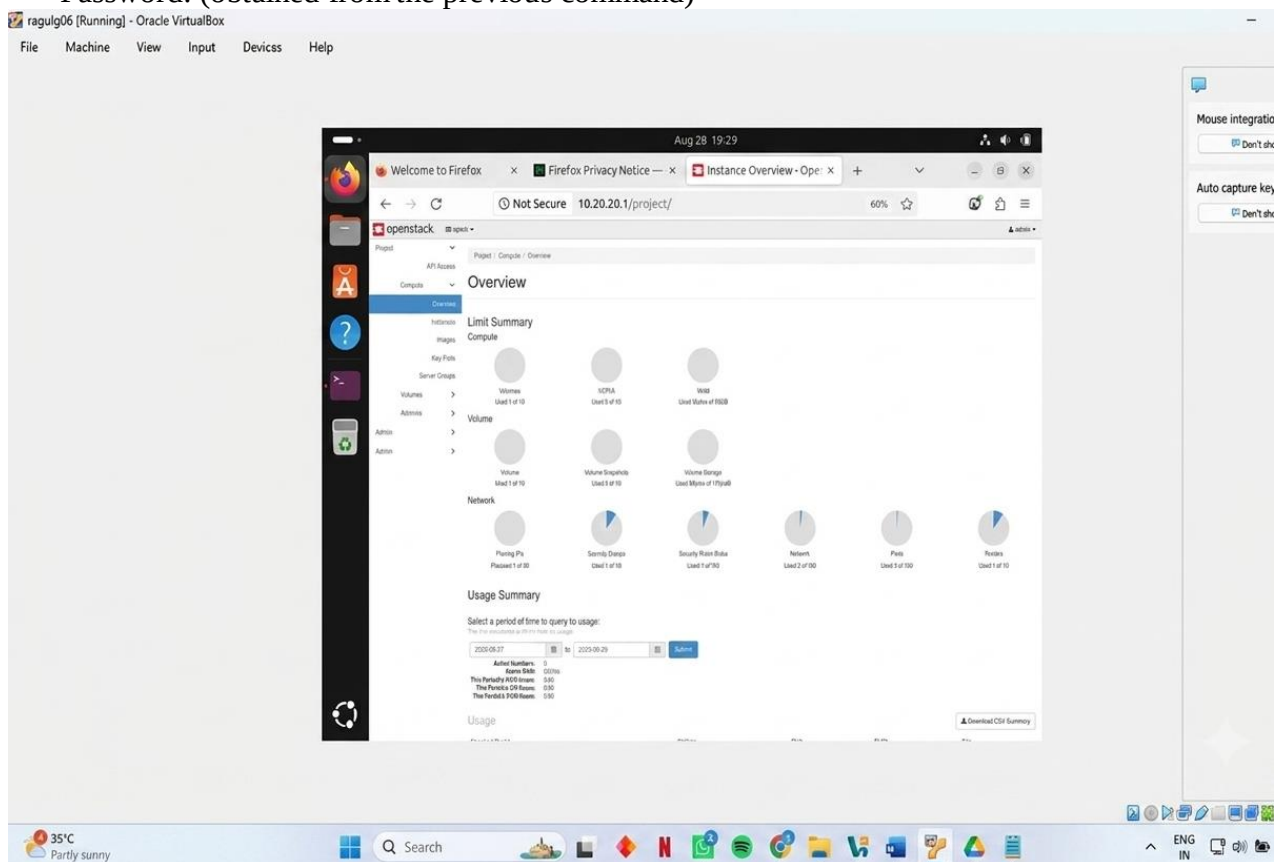
Retrieve the Keystone administrator password:

```
sudo snap get microstack config.credentials.keystone-password
```

Step 9

Open the OpenStack Horizon Dashboard in the web browser using the IP address provided after initialization and log in using:

- Username: admin
- Password: (obtained from the previous command)



CONCLUSION:

OpenStack (MicroStack) was successfully installed and configured on Ubuntu, and the Horizon Dashboard was accessed successfully.